Fraud Detection on Medicare Claims Using SQL-Based Feature Engineering and Machine Learning

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1. Abstract

Healthcare fraud in Medicare always results in substantial financial losses and is getting hard to detect since the claims data are sophisticated and imbalanced. In this paper, we present a complete pipeline combining SQL feature store and supervised binary classification and unsupervised clustering on the Medicare dataset. We performed the unsupervised clustering only on the ground level fraudulent cases, and then we headed over classifying the fraudulent providers and clustered the high-risk providers into the fraud type groups. We cleaned the inpatient, outpatient and beneficiary tables stored in the PostgreSQL database. We also engineered the reusable provider and claim level features for the model training and evaluation. We included rule-based indicators for duplicate claims, upcoding and overcharging along with 30/90/365 days aggregates, distinct beneficiary counts and peer-normalized z-scores. We applied HDBSCAN algorithm for spotting patterns in inpatient and outpatient claims. For visualization, we used PCA to map out the data structure and association rule mining to label the clusters. We went through a bunch of supervised models and logistic regression ended up with PR-AUC around 0.73 and ROC-AUC around 0.94. Overall, computed features along with the supervised models and unsupervised clustering gave us a strong benchmark for evaluation. These also helped characterize different fraud types effectively.

Keywords: Medicare Fraud Detection, Healthcare Insurance Fraud Types, Tree-based ensemble learning, Unsupervised Anomaly Detection, HDBSCAN Fraud Clustering, SQL-feature Engineering, Association Rule Mining.

2. Introduction

Healthcare fraud and abuse take the limited resources away from patient care and impose substantial financial losses in large public insurance program such as Medicare. It is estimated tht nearly 3-10% of total healthcare expenses may be subject to fraud and waste each year which is tens to hundreds of billions of dollars globally and 300 billion USD annually on an average in the US [3], [8], [12]. This waste also eroded the trust in health systems [12], [14]. Reviews show that as the health insurance costs are on the rise in many countries, fraudulent activities are also increased proportionally which harms the patient safety and question the quality of the care [8], [14]. Medicare claims provide a lot of information regarding the patients, providers, diagnoses, procedures, deductibles, reimbursements for detection. However, those features are highly imbalanced and multidimensional [1], [3], [8]. With the multidimensional dataset, the detection is sophisticated since fraud cases are rare among all the claims [3], [8], [12]. Rule-based systems can be implemented easily but there is less room for automation in fraud detection and can generate high false-positive claims [8], [12], [14]. These limitations have inspired comprehensive research in healthcare to invent effective strategies to prevent the fraud [3], [5], [8], [12], [14], [15].

3. Background Studies/Literature Review

Recent studies show variety of audit-based checks and ML techniques for fraud detection in health insurance along with statistical models, ensemble learning, support vector machines, deep learning and hybrid pipelines [3], [8], [12], [14]. Du Preez and others did not find any single best approach while reviewing for machine learning methods [3]. The main reasons are differences in dataset features, different types of fraud and evaluation techniques [3]. Having access to the ground level truth is also a challenge here [3]. Kumaraswamy and others worked on data-mining methods for fraud detection [8]. They focused primarily on the significant financial loss, evolution of fraud schemes, and difference in evaluation between academic models and real-world systems [8]. Thaifur and others came out with the discovery of different types of fraudulent activities, and they focus only on providers [14], they skipped inspections at the claim level. Razzaq and Shah highlighted challenges regarding imbalanced data, scarcity of trustworthy labels, privacy concerns and practicality of the models in usability [12].

Supervised models were used to classify the fraudulent claims and providers. Nabrawi and Alanazi evaluated several supervised models on Saudi insurance claims and their work showed that tree-based models and SVM model worked well if the ground level truth for fraud cases is available [11]. Hamid and others studied the CMS DE-SynPUF claims and their findings showed that C4.5, gradient boosting performed well despite having class imbalance problem and provider-level features are carefully implemented [5]. Liu and Vasarhelyi proposed a clustering model using geographic and payment data [9]. Their approach combined claim and provider features to flag suspicious cases [9]. Kumar and Sountharrajan built optimized deep learning models like CNN with proper hyperparameter tuning for fraud detection and they notably mentioned challenges about the feature-engineering and class imbalance challenges [7].

Unsupervised anomaly detection methods found more suitable when working with limited labels for fraud cases. Bairy and others tested Isolation Forest, Local Outlier Factor, KNN, and autoencoders on healthcare dataset, their methods can spot rare anomalies which need further review [1]. De Meulemeester used deep embeddings to flag anomalies and added necessary explanations for the investigators [2]. Kanksha and others work flagged fraud cases without having enough fraud labels [6]. Settipalli and Gangadharan worked on WMTDBC which converted correlated claim data into independent variables [13]. They also used distance-based detection which improved the detection of multivariate extremities [13]. Massi and others at first grouped providers into peer clusters and then they spotted each of the outliers within the cluster [15]. Saldamli and his team worked on blockchain systems for advanced security in transaction and decentralize the claims data [4]. We addressed these gaps by applying a hybrid approach. As we go along, we combined the SQL-based feature engineering, supervised classification and unsupervised clustering for Medicare claims in this project.

4. Methodology

4.1. Data sources and study design

We used the Medicare dataset from Medical Sphere (Open Source) which comes with 4 training and 4 test files. Both of them have provider, inpatient claims, outpatient claims, and beneficiary tables. The training provider table includes a *PotentialFraud* flag. Inpatient and outpatient tables contain each row with claim IDs, beneficiary IDs, provider IDs, dates, reimbursed amount, deductibles, and diagnosis/procedure codes. The inpatient table also has admission, discharge dates and diagnosis related fields. The beneficiary table contains basic patient information, demographics, and chronic health condition of the patient. We computed mean, median and z-scores on payments, length of stay. We also did the same on total- claims, reimbursement amounts, and unique beneficiaries on 30/90/365 days to classify the fraudulent providers. We utilized claims data for feature construction and performed unsupervised clustering on fraud-only claims. For supervised learning, we trained our models on 70% of the training data, we kept 30% data for evaluation. We then applied the trained models on unlabeled test data to examine risk scores and labels for characterizing the provider-clusters.

4.2 Data cleaning and SQL feature store

All data processing was implemented in PostgreSQL using a layered design. We loaded raw csv files into a staging schema with all fields as text and basic checks on structure and column counts. Cleaned data moved to a curated schema, where placeholder strings for missing values were converted to NULL and columns cast to appropriate types. We went through data processing steps to verify unique provider and beneficiary IDs. We made sure there were non-null values in all the features across the inpatient and outpatient tables. We also looked into valid foreign keys and non-negative payments, length of stay and other numeric features. After handling those checks, we put together a mart schema where we had a fact table in place along with feature tables focused on claims and providers. These tables contain details about the length of stay, code counts for diagnoses and procedures, duplicates, possible overcharges, peer-normalized z-scores and 30/90/365-day aggregates.

4.3 Feature engineering and rule-based indicators

We implemented simple rules on both claim and provider features to capture duplicate billing, upcoding, overcharging, and other fraud patterns. We worked on reimbursed amount, deductible, length of stay, counted the diagnosis and procedure counts depending on the type of the patient. We also detected exact or near duplicates by comparing for the same patient and provider within short span of time. We then calculated peer-normalized z-scores and IQR thresholds for reimbursement payments and grouped patient records across windows of 30, 90, and 365 days. We then observed duplicates, upcoding, and overcharge occurrences to flag suspicious claims and used those inputs for training the models in next steps.

4.4 Exploratory analysis and feature selection

After the initial setup of the SQL feature store was completed, we proceeded with the exploratory analysis and feature selection in Python. With the help of SQLAlchemy and pandas, we loaded the claim and provider level feature tables from the database to inspect the basic distribution of data. To evaluate skewness and long tails for reimbursed amounts, deductibles, and inpatient length of stay, we plotted histograms and boxplots, which confirmed the expected tailed results. We used IQR and median-based z-scores to flag extreme outlier claims and then compared those with overcharge indicators and inpatient claims. We used pairwise correlation coefficients and inflation factors to remove highly collinear features as those are redundant in the feature space. We were interested in the shared ground of recent reimbursement and claim volumes. Therefore, the analysis enabled us to work on the meaningful set of provider features.

4.5 Unsupervised fraud-only claim clustering

We clustered fraud-only claims to understand the diversity in fraudulent behaviors. We at first distinguished the fraudulent providers from the training labels and grouped all the associated claims based on inpatient and outpatient types. We stored those fraudulent records along with the extracted features in separate tables for inpatient and outpatient claims. We looked for missing values and replaced the value with the median of all the records. We scaled features to ensure the contribution to the output equally which also sped up the optimization. It also prevented bias toward the larger numeric features. We then applied HDBSCAN clustering algorithm distinctly to inpatient and outpatient claims. Our goal was to find clusters of different size and density and claims that falling under the minimum number of clusters would be considered as noise. We made sure the minimum size of clusters and samples were tuned for better accuracy. We also visualized the structure via PCA projections using different colors on the clusters. After that, we ran associate rule mining within those clusters to characterize unique types of fraudulent claims.

4.6 Supervised provider fraud classification

We applied supervised learning to classify whether a provider is fraudulent or not based on the collected claim features. We only trained the models on labeled training providers, as we know that these are absolutely fraudulent cases. We split the fraud cases from the train dataset into 70/30 holdout, where we used 70% of the data for training, and 30% data for evaluation. This strategy preserved the proportion between the fraudulent and non-fraudulent claims. We compared the performance using multiple classifiers on the same feature set. We used linear model (logistic regression with elastic-net regularization), tree-based models (Random Forests, Extra Trees, LightGBM, CatBoost, XGBoost) for the benchmark evaluation on accuracy, recall, F1-score, PR-AUC, and ROC-AUC. Based on those models, we used class-weighting or imbalance-aware parameters to address the scarcity of the fraudulent providers. We tuned the hyperparameters using the randomized search with 5-fold cross validation and then optimized the PR-AUC curve. We did not use the fixed 0.5 cutoff threshold as it won't provide always the best result based on the scenario. We chose the probability threshold which gave us the maximum F1-score on the PR-AUC curve. After completing the performance tests, we found logistic regression to be the most performant model and this performance gains mostly have been achieved by the engineered features rather than the model complexity.

4.7 Provider fraud-type clustering

To understand the diversity in fraud patterns, we went beyond classifying a provider fraudulent or not, we tried to characterize the fraud activities among the group of high-risk providers. We characterized on providers already labeled as fraudulent and test providers whose fraud scores have already passed the high threshold. We pulled out the same standardized features used in supervised models. These features were reimbursement amounts, claims volumes across aggregated timeframe, count of distinct patients, peer-normalized inpatient and outpatient z-scores, and rule scores for duplicate, upcoding and overcharging claims. We then applied HDBSCAN algorithm with a minimum cluster size of 10 and a small number of minimum samples value. This formed clusters with varying size and density and few of the providers were scattered across the plot and were considered as noise. We visualized the structure using principal components and colored points by cluster label, with noise in grey. For each cluster, we produced the summary of labeled against high-scored mix and interpreted those fraud types as high-revenue inpatient overchargers, high-volume outpatient overbillers and others.

5. Results

5.1 Data characteristics and exploratory analysis

The training data has 40,474 inpatient claims, 517,737 outpatient claims, 138,556 distinct patients, and 5410 labeled providers along with the ground level information regarding fraud or non-fraud. We found reimbursed amounts highly skewed as expected in the Medicare billing. In most of the cases, the claims had moderate payments, and a small fraction of claims were very costly which produced a low-payment bulk and a long right tail (Figure 1(a)). There were also variations in the inpatient and outpatient claims. Inpatient claims had higher median value on reimbursed amounts and wider spread confirming that inpatient services are more expensive and changeable. In contrast to that, outpatient services show the claims count against the length of stay, diagnosis vs. procedure counts, and reimbursement vs coding intensity. We flagged high-cost claims by using interquartile-range and median absolute deviation rules. We kept a small set of potential outliers as possible fraud and unusual cases. Performing the data preprocessing, we could not find any missing value across the key features. Correlations and variance inflation factors (Figure 2) produced a compact feature set for unsupervised and supervised analyses.

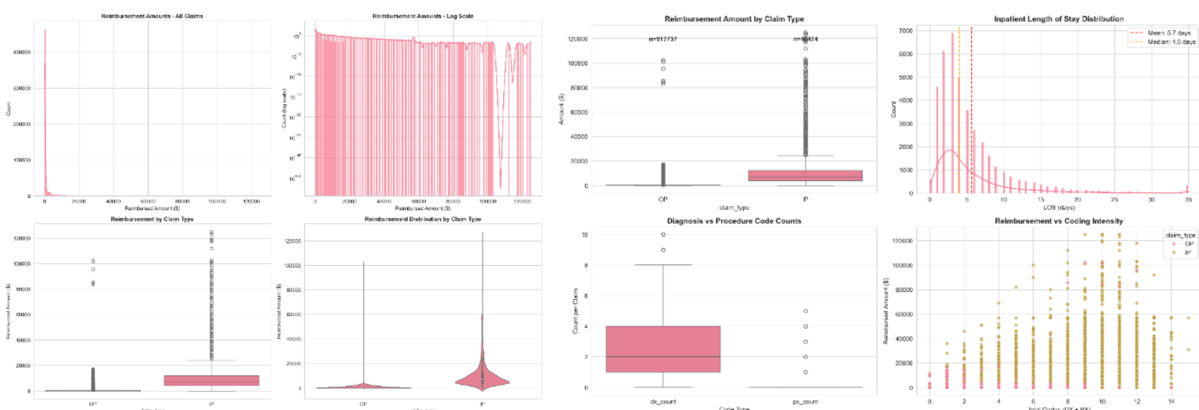

Figure 1. Distribution and comparison of claim reimbursements. (a) Histogram and density plots of reimbursed amounts for inpatient and outpatient claims, showing a heavy right skew and a small number of very high-cost claims. (b) Boxplots of reimbursed amounts and inpatient length of stay, highlighting distributional differences between claim types.

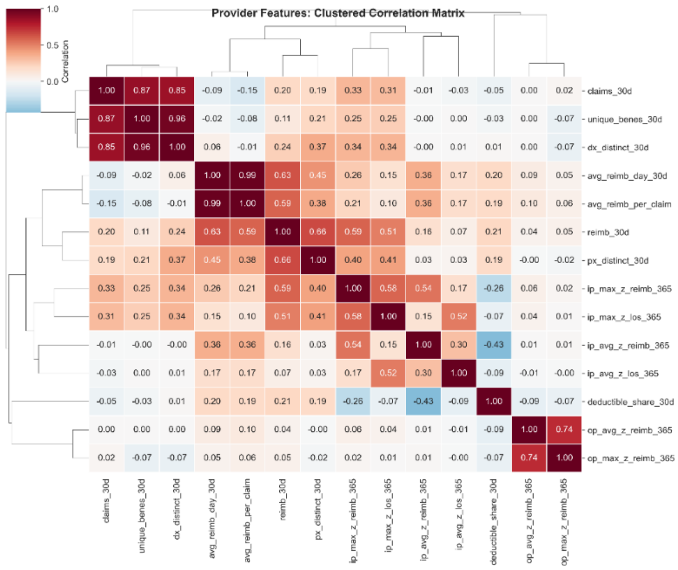

Figure 2. Clustered Pearson correlation heatmap of provider-level features, used to identify highly collinear variables and guide feature selection.

5.2 Unsupervised claim-level patterns among fraudulent claims

To understand the diversity in fraudulent behavior, we only worked on the fraud-only cases. We found out 22,640 inpatient and 8,408 outpatient fraudulent claims out of 558,211 claims. After standardizing the reimbursement amounts, code counts, rule flags, and peer normalized scores, we implemented HDBSCAN to inpatient and outpatient records. HDBSCAN produced several clusters and claims not belonging to any clusters were considered as noise. Figure 3 showed inpatient fraudulent clusters of high-payment with long-stays as we got high reimbursement/length-of-stay z-scores. Figure 4 showed the visualizations of high-payment groups of fraudulent outpatient claims. Most of the outpatient clusters had low to moderate paid claims with a very few higher paid claims. Association rule mining within clusters (Table 1) showed small diagnosis-code sets in outpatient clusters and diagnosis-procedure pairs in high-payment inpatient clusters.

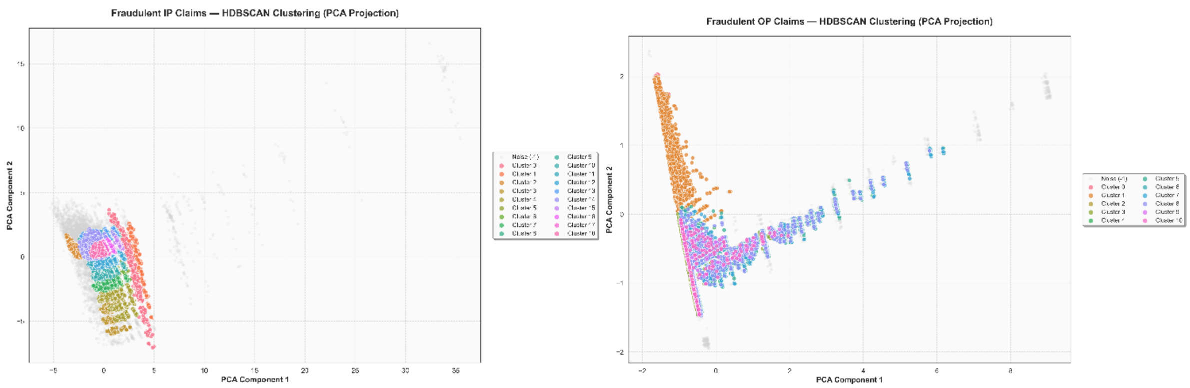

Figure 3–4. Two-dimensional PCA projections of standardized fraud-only claims, showing HDBSCAN clusters of fraudulent inpatient (left) and outpatient (right) claims. Points are colored by cluster labels, with noise points shown in grey.

Claim Type	Cluster	Antecedents Str	Consequents Str	Support	Confidence	Lift
OP	1	280+285	588	0.039	0.766	10.127
OP	1	285+588	280	0.039	0.688	9.462
IP	18	995	38	0.031	0.645	8.896
OP	1	V56	588	0.024	0.637	8.425
OP	1	588	280	0.046	0.612	8.412
OP	1	280	588	0.046	0.636	8.412
IP	14	8154	715	0.031	0.863	7.415
IP	13	7935	820	0.051	0.865	7.398

Table 1. Association rules for diagnosis and procedure code patterns within selected inpatient and outpatient HDBSCAN claim clusters. Results include antecedent code sets, consequent codes, support, confidence, and lift values for fraud-only clusters.

5.3 Provider-level fraud classification performance

We evaluated supervised models on classifying the fraudulent providers among all the cases. On the 30% hold-out evaluation trainset, all models have achieved ROC-AUC around 0.93-0.94 indicating that those models were good at telling apart fraudulent and non-fraudulent providers.

All the performance metrics are reported in Table 2. Logistic regression with elastic-net regularization resulted in the best PR-AUC of around 0.73, ROC-AUC of around 0.94. After threshold tuning, we got precision of around 0.59, recall of around 0.74, and F1-score of around 0.66. Tree-based models (Random Forests, LightGBM, XGBoost, Extra Trees, CatBoost) achieved similar ROC-AUC but weaker PR-AUC and could not surpass logistic regression model (Figure 5). Fraud risk scores using the linear model placed most of the non-fraudulent providers near 0 and fraudulent providers at higher values (Figure 6). Big reimbursements, claim values, z-scores in the peer standard feature sets and rule indicators of duplicates and overcharging contributed towards these high scores mostly. We then applied those scores to the test providers to define the high-risk providers for clustering in Section 5.4.

Model	**Accuracy**	**Precision**	**Recall**	**F1-Score**	**PR-AUC**	**ROC-AUC**
LogReg	0.93	0.59	0.74	0.66	0.73	0.94
ExtraTrees	0.91	0.53	0.74	0.61	0.70	0.93
CatBoost	0.92	0.57	0.64	0.60	0.69	0.93
RandomForest	0.93	0.62	0.61	0.61	0.69	0.93
LightGBM	0.91	0.51	0.78	0.61	0.68	0.93
XGBoost	0.91	0.51	0.78	0.61	0.68	0.93

Table 2. Classification performance of supervised models on the provider hold-out set. Results are reported for logistic regression (elastic net), Random Forests, ExtraTrees, LightGBM, XGBoost, and CatBoost in terms of accuracy, precision, recall, F1 score, PR AUC, and ROC AUC.

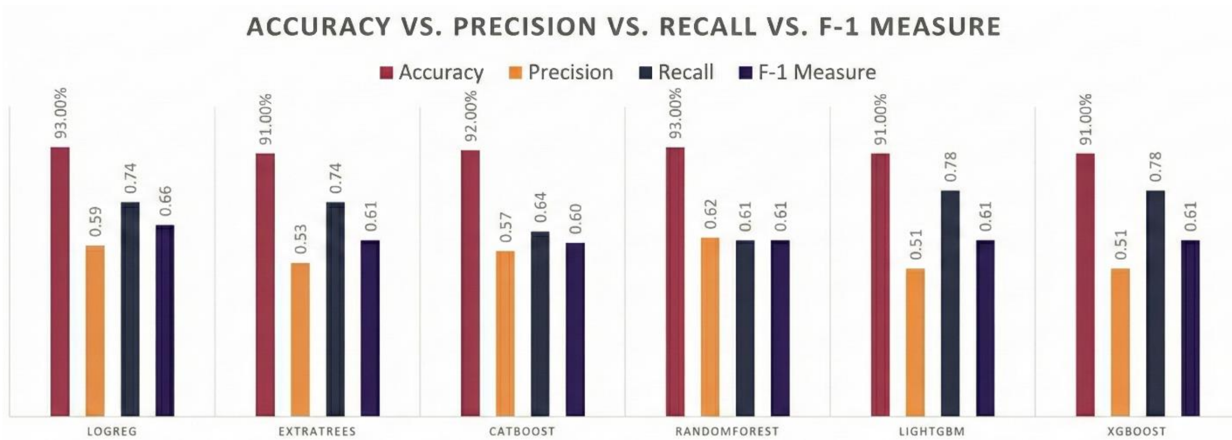

Figure 5. Bar charts of classification performance of supervised models tested for benchmark comparison.

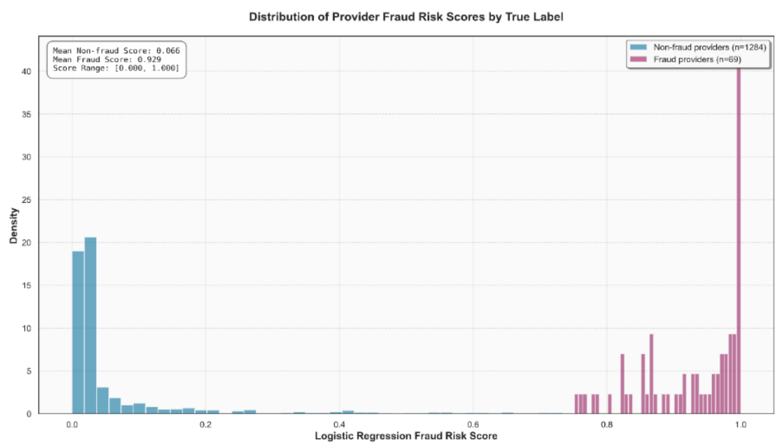

Figure 6. Histogram shows the distribution of logistic regression fraud risk scores for providers. Scores for non-fraudulent providers cluster near zero, while scores for fraudulent providers shift toward higher values.

5.4 Provider fraud-type clusters

Among the high-risk class created of 570 providers, HDBSCAN identified 4 clusters and labeled 334 providers as noise, referring to many high-risk providers identified as noise as no clusters were formed with those points. Figure 7 shows compact regions for clusters with a mix of noise. The largest cluster (Cluster 3) is formed with 180 provider points which refers to the high z-scores of reimbursements and length-of-stay but moderate duplicate and overcharge counts. Cluster 1 (3 providers) shows high revenue providers with high inpatient z-scores, which can be a sign of upcoding. Cluster 0 and Cluster 2 (10 providers each) capture the high count for outpatient billers with duplicates and overcharge flags. Table 3 shows the results of the fraudulent clusters.

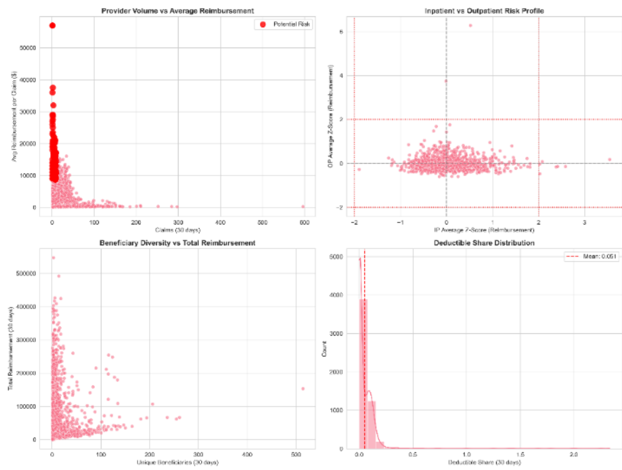

Figure 7. Two-dimensional PCA projection of standardized provider-level features for high-risk providers. Points are colored by HDBSCAN cluster label, with noise providers shown in grey.

cluster	n_providers	n_train_fraud	n_test_highrisk	median_reimb_365d	median_claims_365d
0	10	10	0	252570	309
1	36	36	0	490880	140
2	10	10	0	390030	1336
3	180	180	0	210265	108

Table 3. Provider fraud-type clusters identified by HDBSCAN, showing the number of providers, labeled fraud cases in training, high-scored test providers, and median annual reimbursement and claim volumes.

6. Discussion/Conclusion

In this project we put together a full pipeline for analyzing fraud in Medicare claims data. It involves some feature engineering done with SQL and then classifying providers using supervised methods. We also did unsupervised work to find patterns at the claim level and the provider level too. Our strategy follows recent research combining association rules, supervised models, and unsupervised anomaly detection, rather than relying on a single model [3], [8], [12], [14]. We found that regularized logistic regression model can perform as well as more complex tree-based models. This agrees with other evidence showing that classical methods are still strong for healthcare fraud detection. The performance can be made higher by solid feature engineering [3], [8], [11]. Deep learning provides additional performance gains from the healthcare perspective but requires more computation power and resources [7]. We used HDBSCAN, clustering and association rules to analyze different types of fraud at the claim and provider level. Our approach is similar to current methods in health insurance fraud detection research. Those methods focus on unsupervised learning and clustering across many features [1], [2], [5], [13], [15].

The results suggest interpretable linear models for fraudulent and non-fraudulent provider classification. The unsupervised methods can cluster the high-risk providers and claim-level fraud types effectively. Instead of introducing one algorithm, our work shows how multiple methods can be put together and how multiple algorithms can be used to get a solid performance benchmark. Hence, this pipeline follows current best practices in healthcare fraud research [3], [8], [12], [14].

7. Future Work

We can work more on enhancing the performance of those models by tuning the hyperparameters effectively or focusing on the SQL-feature engineering to raise the performance metrics. Most of the studies confirmed that SQL-feature engineering is one of the crucial steps that play a huge role in raising the evaluation scores [3], [8], [11]. We can also collaborate with one of the healthcare professionals to assess the model's output about flagging the fraudulent cases by validating the association rule mining on those claims. Subject-matter experts can review the top 100 claims or providers from the fraud-type clusters and can help us more with refining the rule definitions. They can also help us to assure which clusters belong to which actionable schemes and provide more reliable labels. Then, we can move on to build real-time fraud scoring system. We will embed the current SQL feature store and trained models into the streaming architecture. New claims will automatically be appended to the staging layer; all the aggregates will be updated near real time. Based on the trained set, the model will assign the fraud scores to the new claims and providers as they arrive. We can also integrate live dashboards where we can see the claims flagged as fraudulent for suspicious behavior and non-fraudulent cases as well. Integrating all the above together would give birth to an innovative system of capturing the atypical claims and providers in the healthcare industry ensuring more transparency.

Author Contributions

Zayed Uddin: conceptualization; resource acquisition; data analysis; project coordination; validation; manuscript preparation (original draft) and revision (review/editing). **Md Mishkat:** conceptualization; validation; drafting of the manuscript and revisions. **Ulugbek Abdurakhmonov:** manuscript drafting; review and revisions.

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